

PHYSICS (Q1. TO Q45.)

- Q1.** A coil of 50 turns and area $1.25 \times 10^{-3} \text{ m}^2$ is pivoted about a vertical diameter in a uniform current of the horizontal magnetic field and carries a current of 2A. When the coil is held with its plane in N-S direction, it experiences a couple of 0.04 N-m; and when its plane is E-W, the corresponding couple is 0.03 N-m. The magnetic induction is
 A) 0.2 T B) 0.3 T
 C) 0.4 T D) 0.5 T
- Q2.** A circuit when connected to an AC source of 12V gives a current of 0.2 A. The same circuit when connected to a DC source of 12V, gives a current of 0.4 A. The circuit is-
 A) Series L R B) Series R C
 C) Series L C D) Series L C R
- Q3.** For a plane electromagnetic wave propagation, the ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is-
 A) $1:C^2$ B) $C:1$
 C) $1:1$ D) $1:C$
- Q4.** An equiconvex lens has power P. It is cut into two symmetric halves by a plane containing the principal axis. The power of one part will be-
 A) $\frac{P}{2}$ B) P
 C) $\frac{P}{4}$ D) Zero
- Q5.** A ray of light travelling in a transparent medium of refractive index μ , falls on a surface separating the medium from air at an angle of incidence of 45° . For which of the following value of μ the ray can undergo total internal reflection -
 A) $\mu = 1.25$ B) $\mu = 1.33$
 C) $\mu = 1.40$ D) $\mu = 1.50$
- Q6.** Two coherent sources of light interfere and produce fringe pattern on screen. For central maximum, the phase difference between the two waves will be -
 A) π
 B) $\frac{\pi}{2}$
 C) $\frac{3\pi}{2}$
 D) Zero
- Q7.** A proton and an alpha particle are accelerated from rest to the same energy. The de-Broglie wavelengths λ_p and λ_α are in the ratio-
 A) $\sqrt{2}:1$
 B) 4:1
 C) 2:1
 D) 1:1
- Q8.** The half-life of a radioactive substance is 30 minutes. The time (in minutes) taken between 40% decays and 85% decay of the same radioactive substance is -
 A) 45
 B) 60
 C) 15
 D) 30
- Q9.** A radioactive nucleus undergoes spontaneous decay in the sequence.

$${}_Z^AX \xrightarrow{\beta^+} {}_{Z-1}B \xrightarrow{\alpha} {}_{Z-3}C \xrightarrow{\beta^-} {}_{Z-2}D$$

 Where Z is atomic number of element X. the possible decay particles in the sequence are-
 A) β^+, α, β^-
 B) β^-, α, β^+
 C) α, β^+, β^-
 D) α, β^-, β^+
- Q10.** The device that can act as a complete electronic circuit is -
 A) Zener Diode
 B) Junction Diode
 C) Integrated Circuit
 D) Junction Transistor

Q11. The electron concentration in n-type semiconductor is the same as hole concentration in a p-type semiconductor. An external electric field is applied across each of them. Compare the current in them-

- A) No current will flow in p-type, current will only flow in n-type
~~B)~~ Current in n-type = current in p-type
 C) Current in p-type > current in n-type
 D) Current in n-type > current in p-type

Q12. The amplitude of modulating signal and High frequency carrier are 10 V and 20 V respectively, then modulation index will be-

- A) $\frac{1}{2}$ ✓ B) $2 \left(\frac{1}{2}\right)$
 C) 30 D) 10

Q13. The unit of thermal conductivity is -

- A) J m K^{-1}
 B) $\text{J m}^{-1} \text{K}^{-1}$
 C) W m K^{-1}
 D) $\text{W m}^{-1} \text{K}^{-1}$ ✓

Q14. Which relation regarding product of two vectors is incorrect?

- A) $\vec{a} \times \vec{a} = 0$
 B) $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$
~~C)~~ $\vec{a} \times \vec{b} = \vec{b} \times \vec{a}$ ✗
 D) $\vec{a} \times \vec{b} = (-\vec{a}) \times (-\vec{b})$

Q15. If energy (E), velocity (V) and time (T) are chosen as fundamental quantities, the dimensional formula of surface tension will be -

- A) $E V^{-1} T^{-2}$
 B) $E V^{-2} T^{-2}$ ✓
 C) $E^{-2} V^{-1} T^{-3}$
 D) $E V^{-2} T^{-1}$

$a = \frac{V}{t}$
 $v = at$
 $v = a \frac{V}{P}$

Q16. A particle moving with velocity $\vec{V}$ is acted by three forces shown by the vector triangle PQR. The velocity of the particle will -

- A) Increase
 B) Decrease
~~C)~~ Remain Constant
 D) Change according to largest force $\vec{PQ}$



Q17. A particle has initial velocity $(3\hat{i} + 4\hat{j})$ and has acceleration $(0.4\hat{i} + 0.3\hat{j})$. Its speed after 10s is -

- A) 10 Units B) 7 Units
~~C)~~ $7\sqrt{2}$ Units D) 8.5 Units

Q18. The motion of a particle along a straight line is described by equation: $x = 8 + 12t - t^3$ Where x in metre and t in second. The retardation of the particle when its velocity becomes zero is -

- A) 6 m/s^2 ~~B)~~ 12 m/s^2
 C) 24 m/s^2 D) Zero

Q19. A car is negotiating a curved road of radius R. The road is banked at an angle θ . The coefficient of friction between the tyres of the car and the road is μ_s . The maximum safe velocity on this road is

- A) $\sqrt{\frac{(\mu_s + \tan \theta)}{(1 - \mu_s \tan \theta)}} \cdot gR^2$
~~B)~~ $\sqrt{\frac{(\mu_s + \tan \theta)}{(1 - \mu_s \tan \theta)}} \cdot \frac{g}{R}$
~~C)~~ $\sqrt{\frac{(\mu_s + \tan \theta)}{(1 - \mu_s \tan \theta)}} \cdot \frac{g}{R^2}$
 D) $\sqrt{\frac{(\mu_s + \tan \theta)}{(1 - \mu_s \tan \theta)}} \cdot gR$

$a \propto d$
 $v = at$
 $v = a \frac{d}{R}$
 $a = \frac{v}{t}$
 $a = \frac{v}{t}$

Q20. The gramophone record is revolving with an angular velocity ω . A coin is placed at a distance r from the centre of record. The static coefficient of friction is μ . The coin will revolve with the record if -

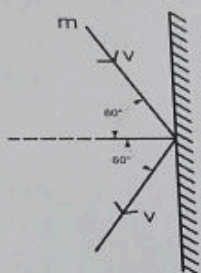
- A) $r \geq \frac{\mu g}{\omega^2}$ B) $r = \mu g \omega^2$
~~C)~~ $r < \frac{\mu g}{\omega^2}$ D) $r \leq \frac{\mu g}{\omega^2}$

- Q21. Two bodies of mass 4 kg and 6 kg are tied to the ends of a mass less string. The string passes over a pulley which is frictionless (see figure). The acceleration of the system in terms of acceleration due to gravity (g) is-



- A) $\frac{g}{10}$ B) $\frac{g}{5}$ C) g D) $\frac{g}{2}$

- Q22. A rigid ball of mass m strikes a rigid wall at 60° and gets reflected without loss of speed as shown in the figure below. The value of Impulse imported by the wall on the ball will be -



- A) $\frac{mV}{2}$ B) $\frac{mV}{3}$ C) mV D) $2mV$

- Q23. What is minimum velocity with which a body of mass m must enter a vertical loop of radius R so that it can complete the loop?

- A) $\sqrt{gR}$ B) $\sqrt{2gR}$ C) $\sqrt{3gR}$ D) $\sqrt{5gR}$

- Q24. In perfect elastic collision in one dimension-

- A) Only Kinetic energy conserved
B) Only Linear momentum conserved
C) Both Kinetic energy and Linear momentum not conserved
D) Both Kinetic energy and Linear momentum conserved

- Q25. A particle covers half of its total distance with speed V_1 and the rest half distance with speed V_2 . Its average speed during the complete journey is -

- A) $\frac{V_1^2 V_2^2}{V_1^2 + V_2^2}$ B) $\frac{V_1 + V_2}{2}$ C) $\frac{V_1 V_2}{V_1 + V_2}$ D) $\frac{2V_1 V_2}{V_1 + V_2}$

- Q26. A solid sphere is rotating freely about its symmetry axis in free space. The radius of sphere is increased keeping its mass same. Which of the following physical quantities would remain constant?

- A) Angular velocity
B) Angular momentum
C) Rotational Kinetic Energy
D) Moment of Inertia

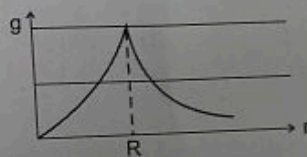
- Q27. Two particles of mass 5 kg and 10 kg respectively are attached to the ends of a rigid rod of length 1 m with negligible mass. The centre of mass of the system from the 5 kg particle is nearly at a distance of-

- A) 67 cm
B) 33 cm
C) 50 cm
D) 80 cm

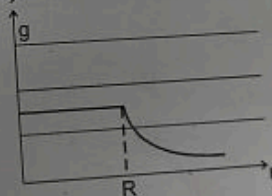


- Q28. Starting from the centre of the Earth having radius R , the variation of g (acceleration due to gravity) is shown by -

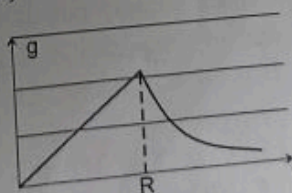
A)



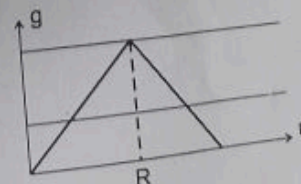
B)



C)



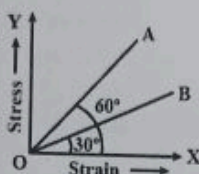
D)



Q29. In a double slit experiment, when light of wavelength 400 nm was used, the angular width of the first minima formed on a screen placed 1 m away, was found to be 0.2° . What will be the angular width of the first minima, if the entire experimental apparatus is immersed in water ($\mu_w = \frac{4}{3}$)

- A) 0.266° B) 0.05°
C) 0.1° D) 0.15°

Q30. The stress versus strain graphs for wires of two materials A and B are as shown in the figure.



If Y_A and Y_B are the Young's moduli of the materials, then:

- A) $Y_B = 2Y_A$ B) $Y_A = Y_B$
C) $Y_B = 3Y_A$ D) $Y_A = 3Y_B$

Q31. A Carnot engine having an efficiency of $\frac{1}{10}$ as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of heat absorbed from the reservoir at lower temperature is -

- A) 99 J B) 90 J ✓
C) 100 J D) 1 J

Q32. The mean square speed of the molecules of a gas at absolute temperature T is proportional to -

- A) $\frac{1}{T}$ B) $\sqrt{T}$
C) T D) T^2

Q33. In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz the original frequency of B will be -

- A) 537 Hz
B) 523 Hz
C) 524 Hz ✓
D) 536 Hz

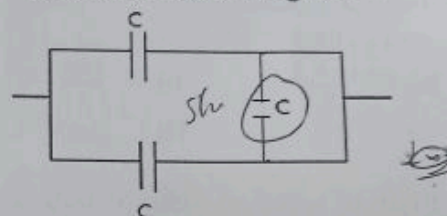
Q34. Average velocity of a particle executing simple Harmonic Motion in one complete oscillation is -

- A) $\frac{A\omega}{2}$ B) $A\omega$
C) $\frac{A\omega^2}{2}$ D) Zero

Q35. The electric field in a region is acting radially outward and is given by $E = Ar$. A charge contained in a sphere of radius 'a' centred at the origin of the field, will be given by -

- A) $A \epsilon_0 a^2$
B) $\epsilon_0 A a^3$
C) $4\pi \epsilon_0 A a^2$ ✓
D) $4\pi \epsilon_0 A a^3$

Q36. The equivalent capacitance of the combination shown in figure is -



- A) C B) 3C
C) 2C ✓ D) $\frac{C}{2}$

Q37. A uniform wire of resistance 50Ω is cut into 5 equal parts. These parts are now connected in parallel. The equivalent resistance of combination is -

- A) 2Ω ✓
B) 10Ω
C) 250Ω
D) 6250Ω

Q38. Consider the following two statements -

i) Kirchhoff's Junction law follows from conservation of charge.

ii) Kirchhoff's loop law follows from conservative nature of Electric field.

- A) Both i) and ii) are wrong
B) Both i) and ii) are correct
C) i) is correct but ii) is wrong ✓
D) ii) is correct but i) is wrong

Q39. A charged particle is moved along a magnetic field line. The magnetic force on the particle is -

- A) Along its velocity
B) Opposite to its velocity
C) Perpendicular to its velocity
D) Zero

$$F = q(v \times B)$$

Q40. The magnetic flux linked with a coil (in Wb) is given by the equation $\phi = 5t^2 + 3t + 16$. The magnitude of induced emf in the coil at the fourth second will be -

- A) 33V
B) 108V
C) 43V
D) 10V

$$\phi = 5t^2 + 3t + 16$$

$$= 10t + 3$$

$$= 43$$

Q41. A parallel plate condenser has a uniform electric field E (v/m) in the space between the plates. If the distance between the plates is d (m) and area of each plate is A (m^2), the energy (Joules) stored in the condenser is

- A) $\frac{E^2 Ad}{\epsilon_0}$
B) $\frac{1}{2} \epsilon_0 E^2 Ad$
C) $\epsilon_0 E A d$
D) $\frac{1}{2} \epsilon_0 E^2$

$$\frac{1}{2} \epsilon_0 E^2 A d$$

Q42. Current sensitivity of a moving coil galvanometer is 5 div / mA and its voltage sensitivity (angular deflection per unit voltage applied) is 20 div / V. The resistance of galvanometer is -

- A) 40 Ω
B) 25 Ω
C) 250 Ω
D) 500 Ω

Q43. Two similar coils of radius R are lying concentrically with their planes at right angles to each other. The currents flowing in them are I and $2I$ respectively. The resultant magnetic field induction at the centre will be

- A) $\frac{\sqrt{5} \mu_0 I}{2R}$
B) $\frac{\mu_0 I}{2R}$
C) $\frac{\mu_0 I}{R}$
D) $\frac{3\mu_0 I}{2R}$

Q44. Two point charges A and B having charges $+Q$ and $-Q$ respectively, are placed at certain distance apart and force acting between them is F . If 25% charge of A is transferred to B, then force between the charges becomes -

- A) F
B) $\frac{9F}{16}$
C) $\frac{16F}{9}$
D) $\frac{4F}{3}$

Q45. In which of the following devices, the eddy current effect is not used?

- A) Induction furnace
B) Electric Heater
C) Electromagnet
D) Magnetic braking in train

A

(A)

$$0.75A$$

B

(B)

$$-1 + 0.25 = -0.75$$

$$0.25 \times 0.75$$

$$\frac{8}{9} \times \frac{3}{4} = \frac{9}{10}$$

(5)

(20)

$$R = \frac{V}{I} = \frac{20 \times 10^3}{5 \times 10^3} = 4000$$

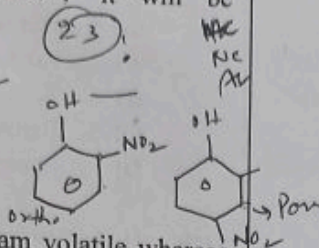
CHEMISTRY (Q46. TO Q90.)

- Q46. Which of the following is the most suitable drying agent for ammonia gas
 A) Calcium Oxide
 B) Anhydrous calcium chloride ✓
 C) Phosphorus Pentoxide
 D) Conc. Sulphuric Acid
- Q47. Lanthanides are the group of elements in which the differentiating electron enters into:
 A) s-sub level
 B) d-sub level
 C) p-sub level
 D) f-sub level ✓
- Q48. An octahedral complex is formed, when hybrid orbitals of the following type are involved
 A) sp^3 B) dsp^2
 C) sp^3d^2 ✓ D) sp^2d
- Q49. Optically active compound is
 A) 2-Chloropropane
 B) 2-Chlorobutane
 C) 3-Chloropentane
 D) None of these ✓
- Q50. $C_6H_5NH_2 \xrightarrow{NaNO_2, HCl} X \xrightarrow{Cu_2(CN)_2} Y \xrightarrow{H_2O/H^+} Z$
 Z is identified as:
 A) $C_6H_5-NH-CH_3$
 B) C_6H_5-COOH ✓
 C) $C_6H_5-CH_2-NH_2$
 D) $C_6H_5-CH_2-COOH$
- Q51. Rosenmund's reduction of an acyl chloride
 A) An aldehyde ✓
 B) An alcohol
 C) An ester
 D) A hydrocarbon
- Q52. The strongest acid among the following aromatic compounds is
 A) *ortho*-nitrophenol
 B) *para*-chlorophenol
 C) *para*-nitrophenol ✓
 D) *meta*-nitrophenol

- Q53. Benzene diazonium chloride on boiling with dilute sulphuric acid gives
 A) Toluene
 B) Benzoic Acid
 C) Benzene
 D) Phenol ✓
- Q54. Benzyl alcohol is obtained from benzaldehyde by
 A) Fittig's reaction
 B) Cannizaro's reaction ✓
 C) Kolbe's reaction
 D) Wurtz's reaction
- Q55. Which of the following is not an amino acid
 A) Glycine
 B) Alanine
 C) Histidine
 D) Benzidine ✓
- Q56. A condensation polymer among the following polymers is
 A) PVC B) Teflon
 C) Decron ✓ D) Polystyrene
- Q57. Which of the following enhances lathering property of soap?
 A) Sodium carbonate
 B) Sodium rosinate ✓
 C) Sodium stearate
 D) Trisodium phosphate
- Q58. Different proportions of oxygen in the various oxides of nitrogen prove the
 A) Equivalent proportion
 B) Multiple proportion ✓
 C) Constant proportion
 D) Conservation of matter
- Q59. Which one of the following explains light both as a stream of particles and as wave motion
 A) Diffraction
 B) $\lambda = h/p$ ✓
 C) Interference
 D) Photoelectric Effect

Q60. If an atom has electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$, it will be placed in

- A) Second Group
B) Third Group
~~C) Fifth Group~~
D) Sixth Group



Q61. Ortho-nitrophenol is steam volatile whereas para-nitrophenol is not. This is due to

- A) Intramolecular hydrogen bonding present in ortho-nitrophenol ✓
B) Intermolecular hydrogen bonding
C) Intramolecular hydrogen bonding present in para-nitrophenol
D) None of these

Q62. A straight chain hydrocarbon has the molecular formula. The hybridisation for the carbon atoms from one end of the chain to the other are respectively $sp^3, sp^2, sp^2, sp^3, sp^2, sp^2, sp$ and sp . The structural formula of the hydrocarbon would be

- A) $\overset{\vee}{\text{CH}_3} - \text{C} \equiv \text{C} - \text{CH}_2 - \text{CH} = \text{CH} - \text{CH} = \text{CH}_2$
 B) $\text{CH}_3 - \text{CH}_2 - \text{CH} = \text{CH} - \text{CH}_2 - \text{C} \equiv \text{C} - \text{CH} = \text{CH}_2$
 C) $\overset{\vee}{\text{CH}_3} - \overset{\vee}{\text{CH}} = \overset{\vee}{\text{CH}} - \overset{\vee}{\text{CH}_2} - \text{C} \equiv \text{C} - \text{CH} = \text{CH}_2$
 D) $\overset{\vee}{\text{CH}_3} - \overset{\vee}{\text{CH}} = \overset{\vee}{\text{CH}} - \overset{\vee}{\text{CH}_2} - \text{CH} = \text{CH} - \text{C} \equiv \text{CH} \checkmark$

Q63. Select one correct statement. In the gas equation, $PV = nRT$

- A) n is the number of molecules of a gas $\propto$
- B) V denotes volume of one mole of the gas $\propto$
- C) n moles of the gas have a volume V ✓
- D) P is the pressure of the gas when only one mole of gas is present ✓

Q64. The correct value of the gas constant R is close to

- A) 0.082 litre-atmosphere K
B) 0.082 litre-atmosphere $K^{-1} \text{ mol}^{-1}$ ✓
C) 0.082 litre-atmosphere $^{-1} K \text{ mole}^{-1}$
D) 0.082 litre $^{-1}$ atmosphere $^{-1} K \text{ mol}$

0.082 litre atmosphere
0.082 Latm K⁻¹ mol⁻¹

$$P U_2 \propto R T$$

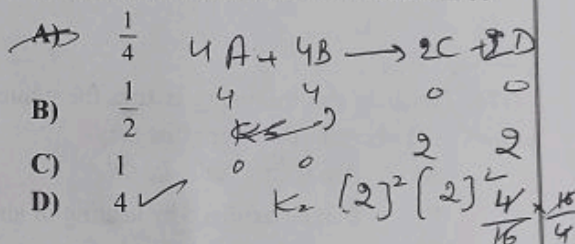
Q65. According to Hess's law

- A) $K_p = K_c \times (RT)^{\Delta n}$
 B) ΔH is always positive
 C) The enthalpy change of a given chemical reaction is same whether the process takes place in one or several steps
 D) None of these

Q66. Which of the following is an extensive property

- A) Heat Capacity ✓
B) Density
C) Surface Tension
D) Specific Heat

Q67. 4 moles of A are mixed with 4 moles of B. At equilibrium for the reaction $A + B \rightleftharpoons C + D$, 2 moles of C and D are formed. The equilibrium constant for the reaction will be



Q68. In the reversible reaction $A + B \rightleftharpoons C + D$, the concentration of each C and D at equilibrium was 0.8 mole/litre, then the equilibrium constant K_c will be

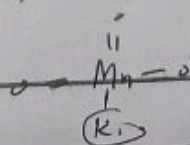
- A) 6.4 B) 0.64 ✓ *0.6 x 0.8*
C) 1.6 D) 16.0

Q69. In which reaction there is a change in valency $+4$ $+4$

- valency
- A) $2\overset{+4}{\text{NO}_2} \rightarrow \overset{+4}{\text{N}_2\text{O}_4}$
- B) $2\overset{+4}{\text{NO}_2} + \text{H}_2\text{O} \rightarrow \overset{+3}{\text{HNO}_2} + \overset{+5}{\text{HNO}_3}$ (0)
- C) $\text{NH}_4\text{OH} \rightarrow \text{NH}_4^+ + \text{OH}^-$ ✓
- D) $\overset{+2}{\text{Ca}}\overset{+4}{\text{CO}_3} \rightarrow \overset{+2}{\text{CaO}} + \overset{+4}{\text{CO}_2}$

Q70. Oxidation number of Mn in K_2MnO_4 and $MnSO_4$ are respectively

- A) $+7, +2$ B) $+6, +2$ ✓
C) $+5, +2$ D) $+2, +6$



SET - D

Q71. Ionization energy of hydrogen is

- A) Equal to that of chlorine
 B) Lesser than that of chlorine
 C) Slightly higher than that of chlorine
 D) Much higher than that of chlorine ✓

Q72. When sodium metal is dissolved in liquid ammonia, blue colour solution is formed. The blue colour is due to

- A) Solvated Na^+ ions
 B) Solvated electrons ✓
 C) Solvated NH_2^- ions
 D) Solvated protons

Q73. The commercial production of sodium carbonate is done by



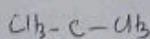
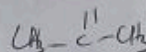
- A) Lead-chamber process
 B) Haber's process ✓
 C) Solvay's process ✓
 D) Castner's process

Q74. Each of the following is true for white and red phosphorus except that they

- A) Are both soluble in CS_2
 B) Can be oxidised by heating in air ✓
 C) Consists of same kind of atoms
 D) Can be converted into one another

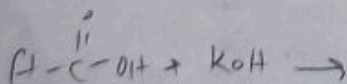
Q75. Compound which gives acetone on ozonolysis

- A) $CH_3-CH=CH-CH_3$
 B) $(CH_3)_2C=C(CH_3)_2$ ✓
 C) $C_6H_5CH=CH_2$
 D) $CH_3CH=CH_2$

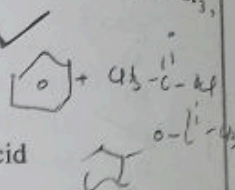


Q76. If formaldehyde and potassium hydroxide are heated, then we get

- A) Acetylene ✓
 B) Methane
 C) Methyl alcohol ✓
 D) Ethyl formate

Q77. When benzene is heated with acetic anhydride in presence of anhydrous $AlCl_3$, the product is

- A) Acetophenone ✓
 B) Toluene
 C) Benzoic acid
 D) Phenyl acetic acid



Q78. Hydrolysis of ozonide of 1-butene gives

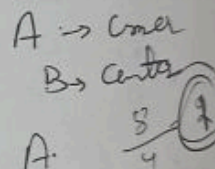
- A) Ethylene only
 B) Acetaldehyde and formaldehyde
 C) Propionaldehyde and formaldehyde ✓
 D) Acetaldehyde and oxalic acid

Q79. A fertile soil is likely to have a pH of

- A) 3
 B) 9
 C) 5.5-7 ✓
 D) 14

Q80. A substance $A_x B_y$ crystallizes in a face centred cubic (FCC) lattice in which atoms A occupy each corner of the cube and atoms B occupy the centre of each face of the cube.The correct composition of the substance $A_x B_y$ is:

- A) AB_3 ✓
 B) A_4B_3
 C) A_3B
 D) None of these



Q81. A non-ideal solution was prepared by mixing 30 ml chloroform and 50 ml acetone. The volume of mixture will be

- A) > 80 ml
 B) < 80 ml ✓
 C) = 80 ml ✗
 D) ≥ 80 ml ✗

Q82. An ideal solution is that which

- A) Shows positive deviation from Raoult's law
 B) Shows negative deviation from Raoult's law
 C) Has no connection with Raoult's law
 D) Obeys Raoult's law ✓

Q83. On passing 0.1 Faraday of electricity through aluminium chloride, the amount of aluminium metal deposited on cathode is ($Al = 27$)

- A) 0.9 gm B) 0.3 gm
C) 0.27 gm D) 2.7 gm ✓

Q84. The rate of reaction is determined by slow step reaction. The step is called

- A) Reaction rate
B) Activation step
C) Rate determining step ✓
D) None of these

Q85. The rate of the reaction

$CCl_3CHO + NO \rightarrow CHCl_3 + NO + CO$ is given by Rate = $K[CCl_3CHO][NO]$. If concentration is expressed in moles/litre, the units of K are

- A) $\text{litre}^2 \text{mole}^{-2} \text{sec}^{-1}$
B) $\text{mole litre}^{-1} \text{sec}^{-1}$ ✓
C) $\text{litre mole}^{-1} \text{sec}^{-1}$
D) sec^{-1} ✓

Q86. In the reaction $2A + B \rightarrow A_2B$, if the concentration of A is doubled and of B is halved, then the rate of the reaction will

- A) Increase by four times
B) Decrease by two times
C) Increase by two times ✓
D) Remain the same

Q87. A substance forms zwitter ion. It can have functional groups

- A) $-NH_2, -COOH$ ✓
B) $-NH_2, -SO_3H$
C) Both (A) and (B) ✓
D) None of these

Q88. Which of the following is contributed towards the extra stability of lyophilic colloids

- A) Hydration ✓
B) Charge
C) Colour
D) Tyndall Effect

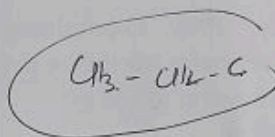
Q89. Which of the following is a tetrabasic acid

- A) Orthophosphorus acid
B) Orthophosphoric acid
C) Metaphosphoric acid ✓
D) Pyrophosphoric acid ✓

Q90. Which salt can be classified as an acid salt

- A) Na_2SO_4
B) $BiOCl$ ✓
C) $Pb(OH)Cl$
D) Na_2HPO_4

$L^{-1} m \cdot l$



$(A)^2 (B)^1$
 $4 \times \frac{1}{2} = 2$

BIOLOGY (Q91. TO Q135.)

- Q91.** HCl in human body is secreted by:
 A) Zymogen cells
 B) Oxyntic cells ✓
 C) Kupfer's cells
 D) Mucous cells
- Q92.** What part of the oviduct does the sperm encounter the egg?
 A) Ampulla B) Isthmus ✓
 C) Fundus D) Infundibulum ✗
- Q93.** What is the stage of the cell cycle at which primary oocytes are arrested in human?
 A) Prophase I ✓ B) Metaphase I ✓
 C) Prophase II D) Metaphase II
- Q94.** Down's syndrome is due to:
 A) Trisomy of chromosome 21 ✓
 B) Trisomy of chromosome 22
 C) Trisomy of X chromosome
 D) Deletion of X chromosome
- Q95.** In forensic study, DNA fingerprinting process is used to determine whether or not an individual is the parent of a child by:
 A) C-DNA analysis
 B) VNTR analysis
 C) Microsatellite analysis ✓
 D) Oligonucleotides assay
- Q96.** Aneuploid females with only one X chromosome is a characteristic of individuals with:
 A) Cri du chat syndrome
 B) Klinefelter syndrome
 C) Familial down syndrome
 D) Turner syndrome ✓
- Q97.** First vertebrates originated during:
 A) Devonian B) Silurian
 C) Carboniferous ✓ D) Ordovician
- Q98.** The point at which funiculus touches the ovule is:
 A) Chalaza B) Hilum ✓
 C) Raphe D) Endothelium
- Q99.** Which of the following is a viral disease?
 A) Tetanus B) Diphtheria ✓
 C) Polio D) All of these
- Q100.** Which of the following antibodies are first produced during the primary immune response?
 A) IgG B) IgA ✓
 C) IgM D) IgE
- Q101.** Who discovered and characterized restriction enzymes in the late 1960s:
 A) Jacob and Monod
 B) Boyer and Cohen ✓
 C) Kary Mullis
 D) Nathan, Arber and Smith
- Q102.** Genetically modified Bt Brinjal has been developed for:
 A) Enhancing shelf life ✗
 B) Drought resistance
 C) Insect resistance ✓
 D) Fungal resistance ✓
- Q103.** The five kingdom classification is based on:
 A) Mode of nutrition
 B) Complexity in cell structure
 C) Complexity of organism's body
 D) All of these ✓
- Q104.** Which is highest in the hierarchy of taxonomic category? *KPC*
 A) Genus B) Class ✓
 C) Family D) Order
- Q105.** In the life cycle of moss, which of the following generation is dominant?
 A) Haploid gametophyte ✗
 B) Diploid gametophyte ✓ *moss*
 C) Haploid sporophyte
 D) Diploid sporophyte *Bryophyte*
- Q106.** Red rust of tea is caused by:
 A) A red alga B) A fungus ✓
 C) A green alga D) A bacterium

- Q107.** Spores in Pteridophytes are:
 A) Haploid ✓ B) Diploid
 C) Triploid D) Tetraploid
- Q108.** In plant lateral root initiates from:
 A) Pericycle ✓ B) Cortex
 C) Endodermis D) Epidermis
- Q109.** Collenchymatous hypodermis is a characteristic of
 A) Dicot stem ✓
 B) Monocot stem
 C) Monocot and Dicot Stem
 D) Hydrophytes
- Q110.** Transpiration is high under:
 A) Dry Environment ✓
 B) High Atmospheric Pressure
 C) Low Temperature
 D) Humid Environment
- Q111.** Method of passive transport which utilizes the special protein called carrier proteins is called:
 A) Facilitated diffusion ✓
 B) Simple diffusion
 C) Active transport
 D) None of these
- Q112.** Where does the Calvin cycle take place?
 A) Thylakoid membrane ✓
 B) Stroma of chloroplast
 C) Cytoplasm
 D) Intermembrane space of chloroplast
- Q113.** The hormone reducing transpiration rate by inducing stomatal closure is:
 A) Auxin B) Cytokinin ✓
 C) ABA D) Ethylene
- Q114.** Generative nucleus in pollen grain of seed plant divides forming:
 A) 2 male nuclei ✓
 B) 2 female nuclei
 C) 1 male nuclei
 D) 1 male nuclei and 1 tube nucleus
- Q115.** If the number of chromosomes in the vegetative tissue of an angiosperm plant is 12, what will be normal number in the endosperm?
 A) 18 ✓ B) 12
 C) 6 D) 24
- Q116.** Transfer of pollen grains from anther to stigma of another flower of the same plant is called:
 A) Geitonogamy B) Xenogamy
 C) Autogamy D) Cleistogamy
- Q117.** The typical Mendelian dihybrid test cross ratio is:
 A) 9:3:3:1 ✓ B) 1:1:1:1
 C) 9:7 D) 1:2:1
- Q118.** At which end are the new DNA bases added during DNA replication?
 A) 5' triphosphate end ✓
 B) 3' triphosphate end
 C) 3' OH end
 D) 5' OH end
- Q119.** A piece of double stranded DNA has 30% A, what will be the % of G?
 A) 30 B) 40 A ✓
 C) 70 D) 20
- Q120.** For food, light and space, the greatest competition is between two:
 A) Closely related species of the same niche ✓
 B) Closely related species of the different niche
 C) Unrelated species of the same niche
 D) Unrelated species of the different niche
- Q121.** Final stage in the tissue culture programme before the new plants are taken out for cultivation in the fields is known as:
 A) Micropropagation
 B) Hardening ✓
 C) Embryogenesis
 D) Explant

- Q122.** Biofortification is the idea of breeding crops to:
- A) make them drought resistant
 - B) make them virus resistant
 - C) increase their economic value
 - D) increase their nutritional value ✓
- Q123.** BOD (biological oxygen demand) is primarily used to:
- A) Estimate the types of microbes
 - B) Estimate the quantity of organic matter in sewage water ✓
 - C) Determine the level of dissolved oxygen
 - D) None of the above
- Q124.** Edaphic factors of ecosystem deal with:
- A) Soil ✓
 - B) Wind
 - C) Rain fall
 - D) Temperature
- Q125.** The speciation in which a population splits into two geographically isolated population experience dissimilar selective pressure and genetic drift is known as:
- A) Sympatric speciation
 - B) Parapatric speciation ✓
 - C) Peripatric speciation
 - D) Allopatric speciation
- Q126.** Reptiles share the following characters with birds and mammals:
- A) Amnion ✓
 - B) Diaphragm
 - C) Homoeothermic
 - D) None
- Q127.** In which form of DNA, the number of base pair per helical turn is 10.5?
- A) B ✓
 - B) Z
 - C) A
 - D) X
- Q128.** Which phase of interphase refers to the synthesis of DNA necessary for division?
- A) G1
 - B) G2
 - C) S ✓
 - D) G0
- Q129.** Golgi apparatus is absent in:
- A) Higher plants
 - B) Liver cells
 - C) Bacteria and blue-green algae ✓
 - D) Yeast
- Q130.** Blood is a specialized:
- A) Epithelial tissue
 - B) Connective tissue ✓
 - C) Elastic tissue
 - D) Nervous tissue
- Q131.** Ligament is a fibrous connective tissue which connects:
- A) Bone to bone ✓
 - B) Muscle to a bone ✓
 - C) Nerve to a muscle
 - D) Skin to a muscle
- Q132.** Sarcomere is an area between two:
- A) H bands
 - B) Z bands ✓
 - C) A bands
 - D) I bands
- Q133.** Heart of man is:
- A) Myogenic ✓
 - B) Neurogenic
 - C) Cardiogenic
 - D) Digenic
- Q134.** Ultrafiltration is the removal of fluid in body which occurs at:
- A) Glomerulus ✓
 - B) Urinary bladder
 - C) Blood capillary
 - D) Tissue fluid
- Q135.** In cockroach alary muscles are associated with:
- A) Tracheae and spiracles ✓
 - B) Hearts and blood circulation ✗
 - C) Feeding and vision ✗
 - D) Wings and locomotion ✓